

Table 1. Full MGSM multilingual-transfer results for Qwen3-4B. Each language uses the same 250 problem identities. Higher is better for every column.

Language	Method	Overall steering (0–2) ↑	Exact answer (%) ↑	Spanish rule (0–2) ↑	Instruction relevance (0–2) ↑	Fluency (0–2) ↑
Chinese	Original	0.000	71.0	0.00	1.80	1.71
	S-PID	0.000	0.0	2.00	0.00	0.00
	A-LQR	1.395	20.0	1.98	1.53	1.15
	H_∞ (ours)	1.708	51.0	2.00	1.87	1.52
French	Original	0.000	60.0	0.00	1.93	1.70
	S-PID	0.000	0.0	1.92	0.00	0.00
	A-LQR	1.388	29.0	1.94	1.62	1.11
	H_∞ (ours)	1.658	48.0	1.96	1.96	1.42
Japanese	Original	0.000	56.0	0.00	1.80	1.64
	S-PID	0.000	0.0	2.00	0.00	0.00
	A-LQR	1.247	32.0	1.84	1.63	1.01
	H_∞ (ours)	1.477	40.0	1.98	1.71	1.20
Swahili	Original	0.000	15.0	0.00	1.43	1.23
	S-PID	0.000	0.0	1.96	0.00	0.00
	A-LQR	0.949	12.0	1.40	1.32	0.97
	H_∞ (ours)	0.612	14.0	0.92	1.44	1.06
Telugu	Original	0.000	6.0	0.00	1.83	1.05
	S-PID	0.000	0.0	2.00	0.00	0.00
	A-LQR	1.369	19.0	1.88	1.46	1.18
	H_∞ (ours)	0.249	8.0	0.34	1.33	1.05